CHAPTER 1 | LOGIC AND PROOFS

Discrete Mathematics • Challenge Test • Question Paper

Time: 150 minutes **Total:** 100 marks **Scope:** Sections 1.1-1.6

Name: _____ **Student ID:** _____

Instructions. Answer all ten questions. Use separate paper if needed. Show your reasoning and name inference rules when requested. All domains are nonempty. OR is inclusive unless stated otherwise. A truth value without its requested justification earns no credit. Equivalent correct formulations are accepted.

1. Translate and analyze system requirements [8 marks]

Let p mean "the request is authenticated," q mean "an override is enabled," and r mean "the request is admitted." The system has four requirements:

R1. A request is admitted only if it is authenticated or an override is enabled.
R2. If an override is enabled, the request is not authenticated.
R3. The request is admitted whenever it is authenticated and no override is enabled.
R4. Enabling an override is sufficient for admitting the request.

(a) [4] Translate R1-R4 into propositional formulas.
(b) [2] Find **all** satisfying assignments, in the order (p,q,r).
(c) [2] Do the four requirements imply $r \leftrightarrow (p \oplus q)$? Justify your answer.

2. Truth table and an exact failure condition [8 marks]

Define $A=(p \oplus q) \rightarrow r$, $B=(p \rightarrow r) \wedge (q \rightarrow r)$, and $F=A \leftrightarrow B$.

(a) [6] Construct an eight-row truth table with columns $p,q,r,p \oplus q,A,p \rightarrow r,q \rightarrow r,B,F$.
(b) [1] Classify F as a tautology, contradiction, or contingency.
(c) [1] Give a simpler formula equivalent to F that uses only $\neg, \vee$. State exactly when F is false.

3. Equivalence and a restricted connective [10 marks]

(a) [6] Without a truth table, prove the following equivalence. Name the laws used at the main steps: $(p \wedge q) \vee (\neg p \wedge r) \vee (q \wedge r) \equiv (p \wedge q) \vee (\neg p \wedge r)$.

(b) [4] Define NOR by $p \downarrow q \equiv \neg(p \vee q)$. Construct a formula equivalent to $p \leftrightarrow q$ using **only** $p,q,\downarrow$, and parentheses. Justify the construction. Repeated subexpressions are allowed; named intermediate formulas may be used in the explanation but not as extra connectives.

WORKING SPACE

4. Quantifier order and a change of domain [8 marks]

For (a)-(c), every variable ranges over $\mathbb{Z}$. Determine each truth value. For a true statement, give witnesses that work for all preceding universal choices. For a false statement, give a counterexample or explain why every proposed witness fails.

(a) [2] $\forall x \exists y \forall z (x+y \leq z^2)$.
(b) [2] $\exists y \forall x \exists z (z^2 < x+y)$.
(c) [2] $\forall x \exists y (x < 2y \wedge 2y < x+2)$.
(d) [2] Re-evaluate (c) when all variables range over $\mathbb{R}$. Justify any change.

5. Restricted uniqueness and negation [12 marks]

The domain consists of all students in a seminar. Let $R(x,y)$ mean "student x reviews student y's report." Self-review is allowed unless a statement explicitly excludes it. Use only R, equality, quantifiers, and logical connectives. Do not use $\exists!$ or introduce new predicates in your final formulas.

(a) [4] Translate: "Every student has exactly one reviewer other than themselves."

(b) [4] Translate: "Exactly one student reviews every other student's report and does not review their own report." Uniqueness applies to this entire property.

(c) [4] Negate your formula in (a). Eliminate every $\rightarrow$ and $\leftrightarrow$, and put every $\neg$ immediately before R. You may use $=$ and $\neq$. Also state the negation in clear English.

6. Finite expansion and countermodels [8 marks]

Let $D=\{a,b,c\}$, where the three elements are distinct, and let R be an arbitrary binary predicate on D. Consider:

A: $\forall x \exists y (x \neq y \wedge R(x,y))$,
B: $\exists y \forall x (x \neq y \rightarrow R(x,y))$.

(a) [3] Expand A into a quantifier-free formula using only $\wedge, \vee$ and atomic expressions $R(u,v)$. Evaluate all comparisons.
(b) [2] Does A imply B? If not, specify a complete countermodel.
(c) [3] Does B imply A? If not, specify a complete countermodel and explain why it works.

For a complete countermodel, list every ordered pair for which R is true; all unlisted pairs are to be false. Do not assume that the two formulas have the same witness requirements.

WORKING SPACE

7. A derivation with two existential witnesses [10 marks]

Let E,T,A,P,Q be unary predicates and M a binary predicate on a nonempty domain. Assume:

I. $\forall x(E(x) \to \exists y(M(y,x) \land T(y)))$
II. $\forall x \forall y((M(y,x) \land T(y)) \to A(x))$
III. $\forall x(A(x) \to (P(x) \lor Q(x)))$
IV. $\exists x(E(x) \land \neg P(x))$

Derive $\exists x(A(x) \land Q(x))$. Give a numbered argument and name each inference rule. Use a fresh name for each existential instantiation; fresh names do not assert that the corresponding objects are distinct.

8. Audit a proof and find a smallest countermodel [12 marks]

A student claims that the following three premises are inconsistent. Here is the proposed proof:

Line	Statement	Claimed reason
1	$\forall x \exists y\ R(x,y)$	Premise I
2	$\forall x \forall y(R(x,y) \to P(y))$	Premise II
3	$\exists x\ \neg P(x)$	Premise III
4	$\exists y\ R(a,y)$	UI, 1; a is arbitrary
5	$R(a,b)$	EI, 4; b is fresh
6	$\forall x\ R(x,b)$	UG, 5
7	$\exists y \forall x\ R(x,y)$	EG, 6
8	$P(b)$	UI and MP, 2 and 5
9	$\forall x\ P(x)$	UG, 8
10	$\neg P(b)$	EI, 3
11	$P(b) \land \neg P(b)$	Conjunction, 8 and 10

(a) [6] Identify every **independent invalid use** of an inference rule and explain the restriction violated. Distinguish a new rule error from a valid rule applied to an unsupported line.
(b) [3] Give a finite model in which all three premises are true but line 7 is false. Specify the domain, the extension of P, and every true pair of R.
(c) [3] Prove that your model in (b) has the smallest possible domain.

WORKING SPACE

9. Parity, contraposition, and impossibility [14 marks]

For integers m,n, define $H(m,n)=m^2+mn+n^2$. You may use elementary parity rules for sums and products, but **do not use modular arithmetic**.

(a) [7] Prove that $H(m,n)$ is even if and only if m and n are both even. Prove the forward direction by contraposition using an exhaustive set of parity cases, and prove the reverse direction directly.

(b) [5] Using (a), prove by contradiction that no integers m,n,k satisfy $m^2+mn+n^2=4k+2$.

(c) [2] Disprove the claim "If $H(m,n)$ is odd, exactly one of m,n is odd." State the correct equivalence.

10. Existence, uniqueness, and contradiction [10 marks]

(a) [6] Prove that for every integer n there is a unique **ordered pair** of integers (q,r) such that $n=2q+r,\ \ r\in\{0,1\}$.
Prove both existence and uniqueness, and make sure your reasoning includes negative integers. Do not cite the division algorithm.

(b) [4] Prove by contradiction that, for every nonzero irrational real number t, at least one of $t+1/t \quad \text{and} \quad t-1/t$ is irrational. You may use the closure of rational numbers under addition and division by a nonzero rational number.

End of question paper. Check that every universal claim has a general argument, every false claim has a complete counterexample, and every uniqueness proof includes existence.

WORKING SPACE